

Final Evaluation: Project 1 - GanForge

Title: Artistic Image Style Transfer using Cycle-GANs

1. Introduction: Style transfer is a fascinating task in computer vision where we take the content of one image and blend it with the artistic style of another. In this project, we employ Cycle-GANs to achieve style transfer without relying on paired image data. We specifically focus on translating ordinary photographs into paintings resembling Claude Monet's iconic style.
2. Motivation: Older style transfer methods required strictly paired datasets, which are hard to find. Cycle-GAN overcomes this limitation by doing unpaired image-to-image translation. This lets us learn an artist's unique style and apply it broadly.
3. Dataset: We utilize the 'monet2photo' dataset provided by the Cycle-GAN creators. It includes 'trainA' (Monet paintings) and 'trainB' (Real photographs), along with testing sets. Since it's unpaired, there are no direct corresponding images.

4. Methodology:

Cycle-GAN Overview: The architecture consists of two Generators (G, F) and two Discriminators (D_A, D_B). It relies on three primary loss functions: Adversarial Loss (for realism), Cycle Consistency Loss (to ensure reverse translation matches the original), and Identity Loss (to maintain color profiles).

5. Results & Observations:

The output images strongly mimic Claude Monet's painting style, adopting his brush strokes and colors while preserving the original photograph's structure. Cycle-GAN proved highly effective for this unpaired translation task, and identity loss helped prevent unwanted color shifts.

6. Conclusion:

We successfully demonstrated unpaired artistic style transfer using Cycle-Consistent Adversarial Networks. Ordinary photographs were transformed into Claude Monet artworks, showcasing the immense potential of GANs in creative media.